

YU-NIEN LIU

Front-End Developer & Creative Technologist

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PROFESSIONAL SUMMARY

Senior-level Front-End Developer and Creative Technologist specializing in React architecture, Web3D (R3F/Three.js), and enterprise-grade AI UI integration. Expert at bridging human instinct and digital intention through cinematic audio-visual storytelling. Proven track record in optimizing high-performance systems and architecting complex global state for mission-critical industrial AI applications.

EDUCATION

Bachelor of Science in Engineering Science

Graduated: 2018

National Cheng Kung University (NCKU) | Tainan, Taiwan

CORE SKILLS

Frontend & Architecture: React, Next.js, ES6+, Redux, GSAP, Framer Motion, HTML5/CSS3

Creative Tech & AI: Three.js, React Three Fiber (R3F), Web Audio API, Transformers.js, GLTF/Draco

Cloud & Infrastructure: Node.js, MongoDB, Firebase (Real-time), RESTful APIs, Git, Vercel/CI/CD

PROFESSIONAL EXPERIENCE

Front-End Developer | Neurelli

Oct 2024 - Feb 2026

- Developed enterprise React applications for AI-driven defect detection, translating high-complexity computer vision models into intuitive, production-ready UI solutions.
- Engineered interactive PyQt camera feed integrations featuring ROI adjustments, dynamic heatmaps, and precise image analysis tools for manufacturing precision.
- Architected global state management using Redux and Redux Persist, ensuring seamless cross-component synchronization and data persistence for long-running sessions.
- Optimized client-side memory handling and migrated real-time monitoring to the browser, reducing server-side overhead by 30% while maintaining 60FPS visual fidelity.

Front-End Developer | Asia Road Technology Co., Ltd.

Jul 2023 - Oct 2024

- Delivered high-performance web solutions and large-scale e-commerce platforms for 30+ corporate clients, transforming complex UI/UX designs into pixel-perfect experiences.
- Spearheaded the restructuring of legacy codebases, significantly improving SEO metrics, accessibility scores, and page load speeds across global deployments.

KEY PROJECTS

Stanly - 3D Narrative Portfolio | stanly-xi.vercel.app

March 2026 - Present

- Engineered a cinematic 3D journey using React Three Fiber (R3F), featuring dynamic voxel evolution and custom post-processing, inspired by Kubrick's "2001: A Space Odyssey."
- Implemented advanced audio logic via Web Audio API, including spatial sound attenuation and seamless transition filters to mirror the visual "evolution" narrative.
- Optimized heavy 3D assets (400MB+) using GLTF/Draco compression and customized asset streaming strategies to achieve rapid initial load times on mobile and desktop.

Entropy Void - Real-time AI Visualizer | charlie-entropy-void.vercel.app

March 2026

- Architected a real-time thermodynamic visualizer using React and Transformers.js for zero-latency, on-device sentiment analysis without backend dependency.
- Integrated Firebase for instant global synchronization and authored custom CSS architectures to bypass native mobile rendering constraints for full-screen immersion.